PID Control of an Under-Damped Second-Order Plant: Effect of Controller Terms on Performance and Robustness

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Abstract. This study examines how proportional, integral and derivative action change the time-domain performance and frequency-domain robustness of a unity-feedback loop around an under-damped second-order plant (ω_n = 10 rad/s, ζ = 0.3, open-loop overshoot 37%). Four controllers built from the same gains (K_p = 3.5, K_i = 6.0, K_d = 0.35) were compared. Proportional control alone gave 64% overshoot, a 22% steady-state error and a 21° phase margin. Adding derivative action raised the phase margin to 85° and reduced overshoot to 20%; adding integral action removed the steady-state error. The full PID controller gave no overshoot, zero steady-state error, an 85° phase margin, infinite gain margin and a 2% settling time of 1.72 s. The settling time is set by a slow closed-loop pole at −1.54 rad/s lying near a controller zero, which leaves a slow approach to the final value.

Index terms: PID control, second-order system, overshoot, settling time, steady-state error, phase margin, sensitivity.

1. Objective and Questions

The objective is to quantify the effect of each PID term on a fixed plant and to assess the resulting loop for both performance and robustness. The questions addressed are:

1. How does proportional action alone affect speed, overshoot and steady-state error?
2. What does integral action contribute, and at what cost in phase margin?
3. What does derivative action contribute to damping and phase margin?
4. What limits the settling time of the combined PID loop?

2. Models

$$G(s) = \omega_n^2 / (s^2 + 2\zeta\omega_n s + \omega_n^2) = 100 / (s^2 + 6s + 100) \quad (1)$$

$$C(s) = K_p + K_i/s + K_d s \quad (2)$$

The plant represents a lightly damped mechanical or electrical system (servomechanism, mass-spring-damper, RLC circuit) [1], [2]. The loop uses unity negative feedback with open-loop transfer function L(s) = C(s)G(s).

3. Method

Step responses and margins were computed for each controller. Time-domain metrics follow the MATLAB *stepinfo* definitions (10% to 90% rise time, 2% settling band). Gain and phase margins are

$$GM = -20 \log_{10}|L(j\omega_{180})|, \quad PM = 180° + \arg L(j\omega_{gc}) \quad (3)$$

Peak sensitivity $M_s = \max|1/(1 + L(j\omega))|$ is also reported, since it bounds both margins ($GM \geq M_s/(M_s - 1)$, $PM \geq 2\arcsin(1/(2M_s))$) [4]. The derivative term is unfiltered, as in Listing A1.

4. Results

Table 1. Simulated closed-loop metrics ($K_p = 3.5$, $K_i = 6.0$, $K_d = 0.35$). Overshoot is relative to the final value.

Controller	Rise time (s)	Settling time (s)	Overshoot	Final value	PM	GM	M_s
P ($K_p = 3.5$)	0.054	1.23	63.8%	0.778	20.8°	∞	2.98
PI	0.063	1.87	38.1%	1.000	16.0°	∞	3.78
PD	0.030	0.26	19.5%	0.778	84.9°	∞	1.00
PID	0.058	1.72	0.0%	1.000	84.7°	∞	1.00

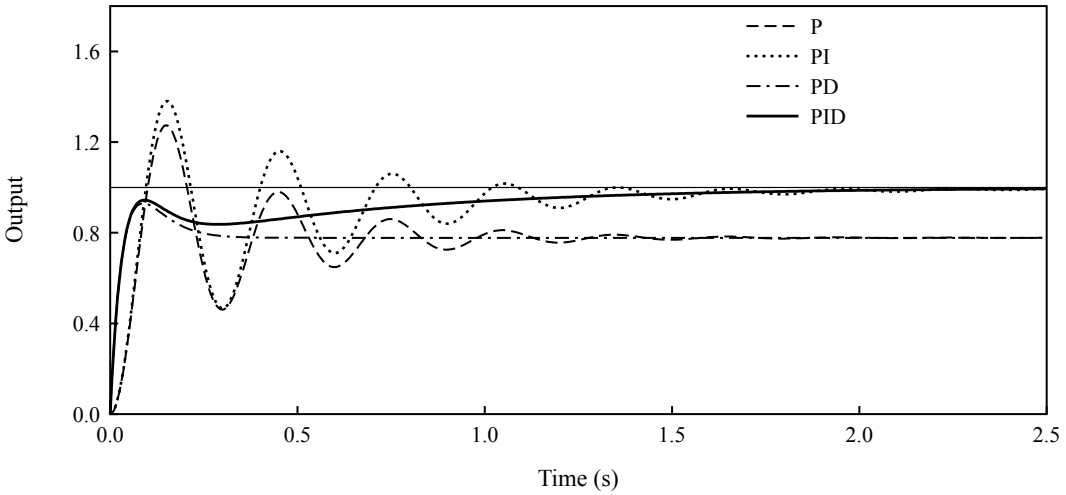


Figure 1. Simulated closed-loop step responses. The thin horizontal line marks the reference.

5. Discussion

Proportional action. $K_p = 3.5$ speeds the response but leaves a steady-state error of $1/(1 + K_p) = 22\%$ for this type-0 plant, and with a 21° phase margin the loop rings strongly.

Integral action. The integrator makes the loop type 1 and removes the steady-state error, but its phase lag reduces the margin from 21° to 16° and raises M_s from 2.98 to 3.78, so overshoot remains high and settling is slower.

Derivative action. Derivative action adds phase lead near crossover, raising the margin to about 85° and reducing overshoot, but does not affect steady-state error.

Combined PID. The controller zeros lie at −2.20 and −7.80 rad/s; the closed-loop poles are at −20.0, −19.5 and −1.54 rad/s. The fast poles give a rapid initial rise (87% of the final value by 0.5 s), while the slow pole near the zero at −2.20 rad/s produces a small, slow tail that sets the 1.72 s settling time. A larger K_i would shorten this tail at the cost of some overshoot and phase margin, which is the main remaining trade-off for this design.

6. Implementation Considerations

- **Derivative filtering.** A practical derivative term requires a filter coefficient N (typically 8 to 20) to limit noise amplification; this slightly reduces phase lead and was not modelled here.
- **Integrator wind-up.** Actuator saturation requires anti-windup (for example, back-calculation).
- **Discretisation.** A digital implementation would use, for example, the Tustin transform with a sample rate of 10 to 20 times the closed-loop bandwidth.
- **Setpoint weighting.** A two-degree-of-freedom structure can shape command and disturbance responses independently.

7. Limitations

- All results are from simulation of a linear model; no physical plant was controlled.
- Only one gain set was evaluated; systematic sweeps of each gain were not performed.
- Actuator saturation, sensor noise and plant-parameter variation were not modelled.

8. Conclusion

For this plant, derivative action provides the phase margin and damping, integral action removes steady-state error, and the combined PID achieves zero overshoot and zero steady-state error with large stability margins. Its settling time is limited by a slow pole-zero pair rather than by oscillation. Future work: sweep K_p, K_i and K_d to map the performance-robustness trade-off, include a derivative filter and actuator limits, and validate on a physical motor.

Appendix A. Analysis Script

```
wn = 10; zeta = 0.3;
G = tf(wn^2, [1 2*zeta*wn wn^2]);
C = pid(3.5, 6.0, 0.35);            % parallel form, unfiltered derivative
T = feedback(C*G, 1);
info = stepinfo(T);                 % rise time, settling time (2%), overshoot
[Gm, Pm] = margin(C*G);             % gain and phase margins
step(T); grid on
```

Listing A1. MATLAB analysis for the PID case.

References

[1] K. Ogata, *Modern Control Engineering*, 5th ed. Pearson, 2010.

[2] N. S. Nise, *Control Systems Engineering*, 8th ed. Wiley, 2019.

[3] G. F. Franklin, J. D. Powell and A. Emami-Naeini, *Feedback Control of Dynamic Systems*, 8th ed. Pearson, 2019.

[4] K. J. Åström and T. Hägglund, *PID Controllers: Theory, Design, and Tuning*, 2nd ed. ISA, 1995.

[5] The MathWorks, "Control System Toolbox: pid, feedback, stepinfo, margin," documentation.